

Cultural Homophily in Grand Slam Doubles

ATP Men's Grand Slam Doubles, 2018–2025 · 1,793 matches after dropping 47 retirements / walkovers · Regression sample: 1,599 GS matches, 3,198 team-obs (complete ranking and nationality/language for all 4 players)

1. Data and Measures

Three culture measures capture teammate similarity. **Same nationality** and **same official language** are binary (0/1). **Language proximity** is a continuous 0–1 index of ethnolinguistic closeness (higher = more similar language family). All descriptive figures show the share of teams (pooled across winners and losers) that are *same* vs. *different* on each dimension.

Regressions use a team-level panel with tournament-by-year fixed effects, round fixed effects, and standard errors clustered by match. Controls: team average doubles ranking, opponent average ranking, teammate rank gap, top-100 singles indicator (either player). Culture measures enter one at a time. Tables 4b and 5b add a team-average GS doubles experience variable (number of prior GS doubles appearances, averaged across both players) and its square as robustness checks. Matches are excluded if either team lacks a valid doubles ranking or if any player's nationality / language data is missing after all imputation (138 matches dropped: 14 for ranking, 124 for nationality / language). Where a player's registered country is unavailable, birthplace country is used as a fallback before coding any remaining unknowns as missing (NaN, not zero).

2. Observation Breakdown

The table below shows how the raw scraped data reduce to the regression sample used in Tables 3–5.

Step	Description	Matches	Change	Team-obs
1	Raw scraped dataset (2018–2025, excl. 2020)	1,840	—	3,680
2	Drop retirements / walkovers (20 walkovers · 9 retired in set 1 · 18 retired in set 2+)	1,793	–47	3,586
3	Drop matches: doubles ranking incomplete for ≥1 player	1,779	–14	3,558
4	Drop matches: nationality / language missing for ≥1 player (birthplace-country used as fallback before this step)	1,655	–124	3,310
5	Exclude Olympic matches (Tokyo 2021 · Paris 2024)	1,599	–56	3,198
→ Grand Slams regression sample (Tables 3–5)		1,599		3,198

Regression sub-samples: Table 4 (regular tiebreaks, sets 1–2) — 712 matches, 1,424 team-obs. Table 5 (adversity: all teams that lost set 1) — 1,599 obs, one per match.

3. Pressure Outcome Counts

Outcome	N	Share of matches
Set-1 tiebreak (7-pt)	445	24.8%
Set-2 tiebreak (7-pt)	470	26.2%
Any regular tiebreak (sets 1–2)	804	44.8%
Match tiebreak / super-tb (10-pt, set 3)	24	1.3%
Any tiebreak (all types)	896	50.0%
Match went to 3 sets	787	43.9%
Comeback wins (winner lost set 1)	351	19.6%

All matches (GS + Olympics, N=1,793). Match tiebreaks concentrated in Olympics (N=20) and Wimbledon post-2018 (N=4).

4. Olympic Cycle Descriptive Evidence

Table 2A — Tokyo 2021 cycle (Grand Slams only, Olympics excluded)

Period	GS window	N teams	Same Nat.	Diff Nat.	Same Lang.	Diff Lang.	Same Ling. Prox.	Diff Ling. Prox.
Pre-Tokyo	AO18 – USO19	1,006	40.1%	59.9%	55.1%	44.9%	55.9%	44.1%
Tokyo Prep	AO20 – Wim21 (ex Wim20)	364	42.0%	58.0%	56.1%	43.9%	56.1%	43.9%
Post-Tokyo	USO21 – USO22	604	39.2%	60.8%	54.8%	45.2%	55.5%	44.5%

N teams = GS matches × 2. Same/Diff = share of all teams (pooled winner + loser) on each cultural dimension.

Table 2B — Paris 2024 cycle (Grand Slams only, Olympics excluded)

Period	GS window	N teams	Same Nat.	Diff Nat.	Same Lang.	Diff Lang.	Same Ling. Prox.	Diff Ling. Prox.
Pre-Paris	AO22 – USO22	486	39.3%	60.7%	54.2%	45.8%	55.1%	44.9%
Paris Prep	AO23 – Wim24	860	42.5%	57.5%	59.1%	40.9%	59.6%	40.4%
Post-Paris	USO24 – USO25	626	45.3%	54.7%	57.7%	42.3%	59.1%	40.9%

N teams = GS matches × 2. Same/Diff = share of all teams (pooled winner + loser) on each cultural dimension.

Cultural Homophily in Grand Slam Doubles – Tokyo 2021 Cycle (Same vs. Diff, by Tournament-Year)

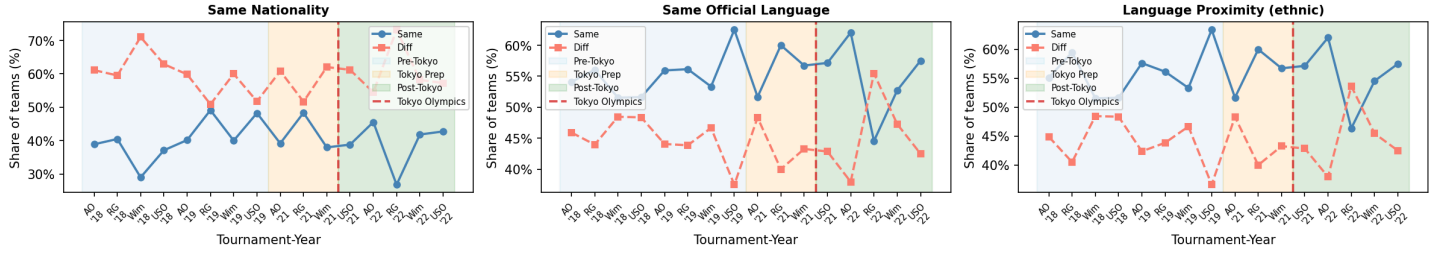


Figure 1. Share of teams that are same (solid) vs. different (dashed) on each cultural dimension, Grand Slams only. Shaded regions: Pre-Tokyo (blue), Tokyo Prep (orange), Post-Tokyo (green). Red dashed line marks the Tokyo Olympics (July–August 2021).

Cultural Homophily in Grand Slam Doubles – Paris 2024 Cycle (Same vs. Diff, by Tournament-Year)

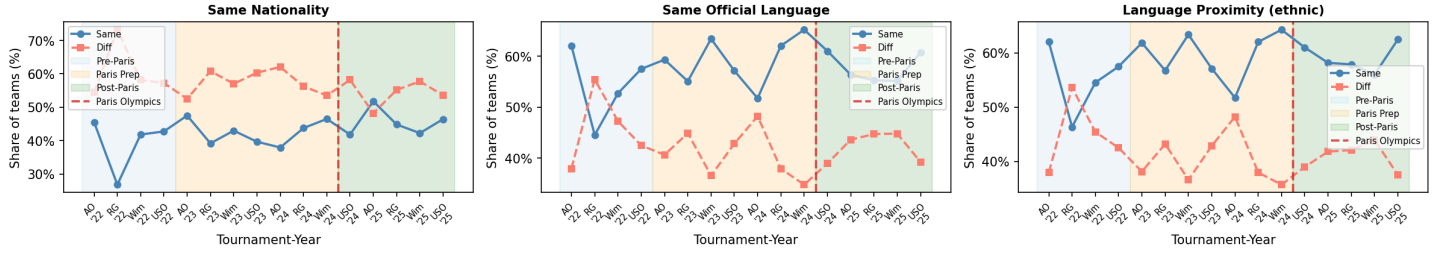


Figure 2. Share of teams that are same (solid) vs. different (dashed) on each cultural dimension, Grand Slams only. Shaded regions: Pre-Paris (blue), Paris Prep (orange), Post-Paris (green). Red dashed line marks the Paris Olympics (July–August 2024).

Cultural Homophily in Grand Slam Doubles – Full Timeline 2018–2025 (Tokyo & Paris Olympic Cycles Combined, Same vs. Diff)

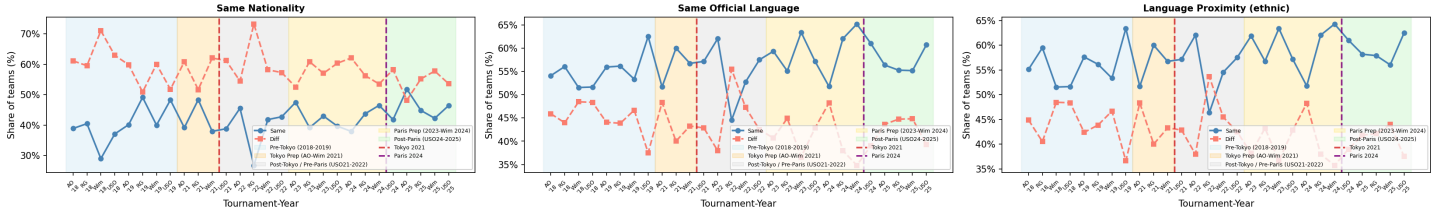


Figure 3. Both Olympic cycles combined. Share of same-culture teams across all Grand Slam tournaments (2018–2025, Olympics excluded), plotted on a single timeline. Shaded regions distinguish preparation and post-Games windows for each cycle; vertical dashed lines mark the Tokyo 2021 and Paris 2024 Olympics.

5. Regression Results

Logit coefficients (log-odds). Each column is a separate model. Standard errors in parentheses, clustered by match. Fixed effects: tournament $\times$ year and round (stage code). Sample: Grand Slams only, excluding matches with incomplete ranking or nationality / language data (N = 1,599 matches, 3,198 team-obs; see Section 2).

Table 3 — Outcome: Match Win | N = 3,198 team-obs (1,599 matches)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.0577	+0.1163	+0.0968
	(0.0791)	(0.0765)	(0.0765)
	p=0.466	p=0.129	p=0.205
Team avg. doubles rank	−0.0041***	−0.0041***	−0.0041***
	(0.0006)	(0.0006)	(0.0006)
Opponent avg. doubles rank	+0.0033***	+0.0033***	+0.0033***
	(0.0003)	(0.0003)	(0.0003)
Teammate rank gap	+0.0008	+0.0008	+0.0008
	(0.0005)	(0.0005)	(0.0005)
Top-100 singles player	+0.0825	+0.0823	+0.0840
	(0.1515)	(0.1518)	(0.1517)
Pseudo-R ²	0.092	0.093	0.093

*** p<0.01 ** p<0.05 * p<0.10. Standard errors clustered by match. FE: tournament × year, round. No culture measure is statistically significant at conventional levels. Sample means (rank_mean | opp_rank_mean | rank_gap): winners = 83.4 | 171.0 | 87.5; losers = 171.0 | 83.4 | 87.5; combined = 127.2 | 127.2 | 87.5. The combined rank means are identical by construction: the panel has two symmetric rows per match where opp_rank_mean is rank_mean of the other row, so their sample averages must be equal.

Table 3b — Match Win + Experience Controls | N = 3,001 team-obs (1,588 matches)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.1031 (0.0833) p=0.216	+0.1613 ^{**} (0.0799) p=0.043	+0.1372 [*] (0.0797) p=0.085
Team avg. doubles rank	−0.0040 ^{***} (0.0007)	−0.0039 ^{***} (0.0007)	−0.0039 ^{***} (0.0007)
Opponent avg. doubles rank	+0.0032 ^{***} (0.0003)	+0.0032 ^{***} (0.0003)	+0.0032 ^{***} (0.0003)
Teammate rank gap	+0.0007 (0.0005)	+0.0007 (0.0005)	+0.0007 (0.0005)
Top-100 singles player	+0.0676 (0.1565)	+0.0673 (0.1568)	+0.0705 (0.1566)
Team avg GS appearances	+0.0792 [*] (0.0440)	+0.0827 [*] (0.0439)	+0.0805 [*] (0.0439)
GS experience squared	−0.0024 (0.0016)	−0.0025 (0.0016)	−0.0025 (0.0016)
Pseudo-R ²	0.093	0.094	0.093

*** p<0.01 ** p<0.05 * p<0.10. Standard errors clustered by match. FE: tournament × year, round. Full GS sample with non-missing experience data (N = 3,001 obs, 1,588 matches; 197 obs dropped for missing experience vs. 3,198 in Table 3). Experience = team average of GS doubles appearances up to current match. Same language and language proximity become marginally significant once experience is controlled for; experience itself is weakly positive (p ≈ 0.06–0.09) with a negative but insignificant quadratic.

Table 4 — Outcome: Match Win (tiebreak matches) | N = 1,594 team-obs (797 matches)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.0868	+0.1324	+0.1258
	(0.1126) p=0.441	(0.1077) p=0.219	(0.1076) p=0.242
Team avg. doubles rank	−0.0023 ^{***}	−0.0023 ^{***}	−0.0023 ^{***}
	(0.0007)	(0.0007)	(0.0007)
Opponent avg. doubles rank	+0.0022 ^{***}	+0.0022 ^{***}	+0.0022 ^{***}
	(0.0004)	(0.0004)	(0.0004)
Teammate rank gap	−0.0002	−0.0002	−0.0002
	(0.0006)	(0.0006)	(0.0006)
Top-100 singles player	+0.3780 [*]	+0.3784 [*]	+0.3796 [*]
	(0.2200)	(0.2206)	(0.2205)
Pseudo-R ²	0.038	0.039	0.039

*** p<0.01 ** p<0.05 * p<0.10. Standard errors clustered by match. FE: tournament × year, round. Sample: all GS matches where any tiebreak occurred (7-pt in sets 1 or 2, or 10-pt match tiebreak). Outcome: match win. Each match contributes two observations: winner (win=1) and loser (win=0). Of the 1,599 baseline GS matches, 797 (49.8%) had at least one tiebreak → 1,594 obs. Sample means (rank_mean | opp_rank_mean | rank_gap): match winners = 84.6 | 134.1 | 51.9; match losers = 134.1 | 84.6 | 95.7; combined = 109.3 | 109.3 | 73.8. Means are symmetric by construction (one winner and one loser per match).

Table 4b — Match Win in Tiebreak Matches + Experience Controls | N = 1,481 team-obs (789 matches)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.1365 (0.1195) p=0.254	+0.1857 (0.1134) p=0.101	+0.1747 (0.1134) p=0.123
Team avg. doubles rank	−0.0024 ^{***} (0.0008)	−0.0023 ^{***} (0.0008)	−0.0023 ^{***} (0.0008)
Opponent avg. doubles rank	+0.0020 ^{***} (0.0004)	+0.0020 ^{***} (0.0004)	+0.0020 ^{***} (0.0004)
Teammate rank gap	−0.0001 (0.0006)	−0.0001 (0.0006)	−0.0001 (0.0006)
Top-100 singles player	+0.3899 [*] (0.2267)	+0.3911 [*] (0.2271)	+0.3934 [*] (0.2269)
Team avg GS appearances	+0.0560 (0.0603)	+0.0596 (0.0601)	+0.0577 (0.0601)
GS experience squared	−0.0013 (0.0022)	−0.0014 (0.0022)	−0.0014 (0.0022)
Pseudo-R ²	0.041	0.042	0.042

*** p<0.01 ** p<0.05 * p<0.10. Standard errors clustered by match. FE: tournament × year, round. Sample: any-tiebreak matches with non-missing experience data (N = 1,481 obs, 789 matches; 113 obs dropped for missing experience). Experience = team average of GS doubles appearances up to current match. Neither experience nor its square is statistically significant; culture coefficients are unchanged.

Table 4c — Match Win, Last-Set Tiebreak Only | N = 374 team-obs (187 matches)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.1105 (0.2513) p=0.660	+0.0882 (0.2349) p=0.707	+0.0636 (0.2342) p=0.786
Team avg. doubles rank	−0.0013 (0.0016)	−0.0012 (0.0016)	−0.0012 (0.0016)
Opponent avg. doubles rank	+0.0012 (0.0008)	+0.0011 (0.0008)	+0.0011 (0.0008)
Teammate rank gap	−0.0006 (0.0014)	−0.0006 (0.0014)	−0.0007 (0.0014)
Top-100 singles player	+1.2695 ^{**} (0.5782)	+1.2680 ^{**} (0.5796)	+1.2660 ^{**} (0.5787)
Pseudo-R ²	0.022	0.022	0.022

*** p<0.01 ** p<0.05 * p<0.10. Standard errors clustered by match. FE: tournament × year, round. Sample: GS matches decided by a last-set tiebreak (set-3 regular 7-pt TB: 183 matches; 10-pt match TB: 4 matches). Compare Table 4 (any tiebreak, 797 matches). Culture coefficients are smaller, statistically indistinguishable from zero, and SEs are roughly twice as large — consistent with low power in a sample one-quarter the size. Top-100 singles player becomes strongly significant in this high-pressure subsample.

Table 5 — Outcome: Comeback Win (Adversity Sample) | N = 1,599 obs (1,599 matches, 1 obs per match)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	−0.2371 [*] (0.1365) p=0.083	−0.3257 ^{**} (0.1324) p=0.014	−0.3316 ^{**} (0.1327) p=0.012
Team avg. doubles rank	−0.0027 ^{***} (0.0010)	−0.0028 ^{***} (0.0010)	−0.0028 ^{***} (0.0010)
Opponent avg. doubles rank	+0.0020 ^{***} (0.0004)	+0.0021 ^{***} (0.0004)	+0.0021 ^{***} (0.0004)
Teammate rank gap	+0.0001 (0.0008)	+0.0001 (0.0008)	+0.0001 (0.0008)
Top-100 singles player	−0.2018 (0.2626)	−0.2015 (0.2628)	−0.2053 (0.2627)
Pseudo-R ²	0.065	0.067	0.067

*** p<0.01 ** p<0.05 * p<0.10. HC3 heteroskedasticity-robust standard errors (one observation per match, no within-match clustering). FE: tournament × year, round. Every match has exactly one team that lost set 1, so the adversity sample spans all 1,599 baseline matches (N = 1,599).

obs = 1 per match). Outcome breakdown: comeback win (0|1|1, N=310, 19.4%), 3-set loss (0|1|0, N=392, 24.5%), straight-set loss (0|0, N=897, 56.1%). Tournament × year cells with no outcome variation dropped. Sample means (rank_mean | opp_rank_mean | rank_gap): comeback wins (win=1) = 101.5 | 121.6 | 118.4; other losses (win=0) = 182.9 | 79.1 | 118.4; combined = 167.1 | 87.3 | 118.4. Unlike Tables 3–4, the two combined rank means differ because only one team per match is kept (the one that lost set 1), breaking the panel symmetry. That team tends to be the weaker side, hence its average rank (167.1) exceeds its opponent's (87.3). Within the comeback sub-group (win=1) the gap shrinks considerably (101.5 vs 121.6), consistent with homophily potentially helping weaker teams recover from adversity.

Table 5b — Comeback Win + Experience Controls | N = 1,483 obs (non-missing experience)

Variable	(1) Same Nationality	(2) Same Language [†]	(3) Lang. Proximity
Culture measure	−0.1912 (0.1443) p=0.185	−0.2469 [*] (0.1370) p=0.072	−0.2734 ^{**} (0.1384) p=0.048
Team avg. doubles rank	−0.0025 ^{**} (0.0011)	−0.0025 ^{**} (0.0011)	−0.0026 ^{**} (0.0011)
Opponent avg. doubles rank	+0.0020 ^{***} (0.0004)	+0.0020 ^{***} (0.0004)	+0.0020 ^{***} (0.0004)
Teammate rank gap	+0.0000 (0.0008)	+0.0001 (0.0008)	+0.0001 (0.0008)
Top-100 singles player	−0.2242 (0.2640)	−0.2116 (0.2620)	−0.2266 (0.2638)
Team avg GS appearances	−0.0200 (0.0693)	−0.0198 (0.0687)	−0.0211 (0.0688)
GS experience squared	+0.0009 (0.0025)	+0.0009 (0.0025)	+0.0009 (0.0025)
Pseudo-R ²	0.062	0.060	0.064

*** p<0.01 ** p<0.05 * p<0.10. HC3 heteroskedasticity-robust standard errors. FE: tournament × year, round. Sample: adversity obs (lost set 1) with non-missing experience data (N = 1,483 obs; 116 dropped for missing experience). [†]Column (2) drops round FE due to near-perfect collinearity in the reduced sample; FE: tournament × year only. Experience controls are statistically insignificant and do not affect culture coefficients materially.